

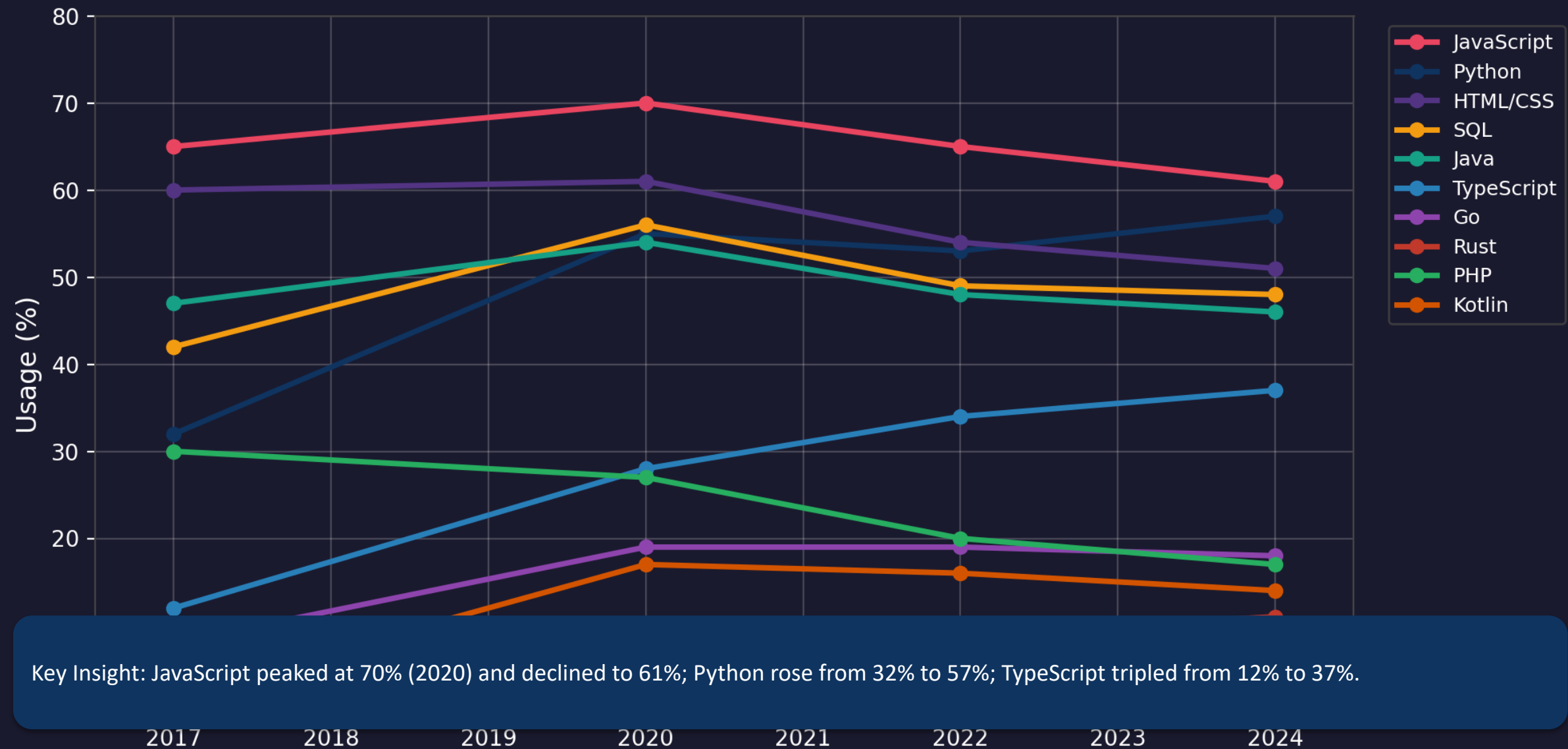
State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

JetBrains State of Developer Ecosystem Report 2024 | 23,262 developers surveyed worldwide

Python Closes the Gap: 8 Years of Language Usage Shifts

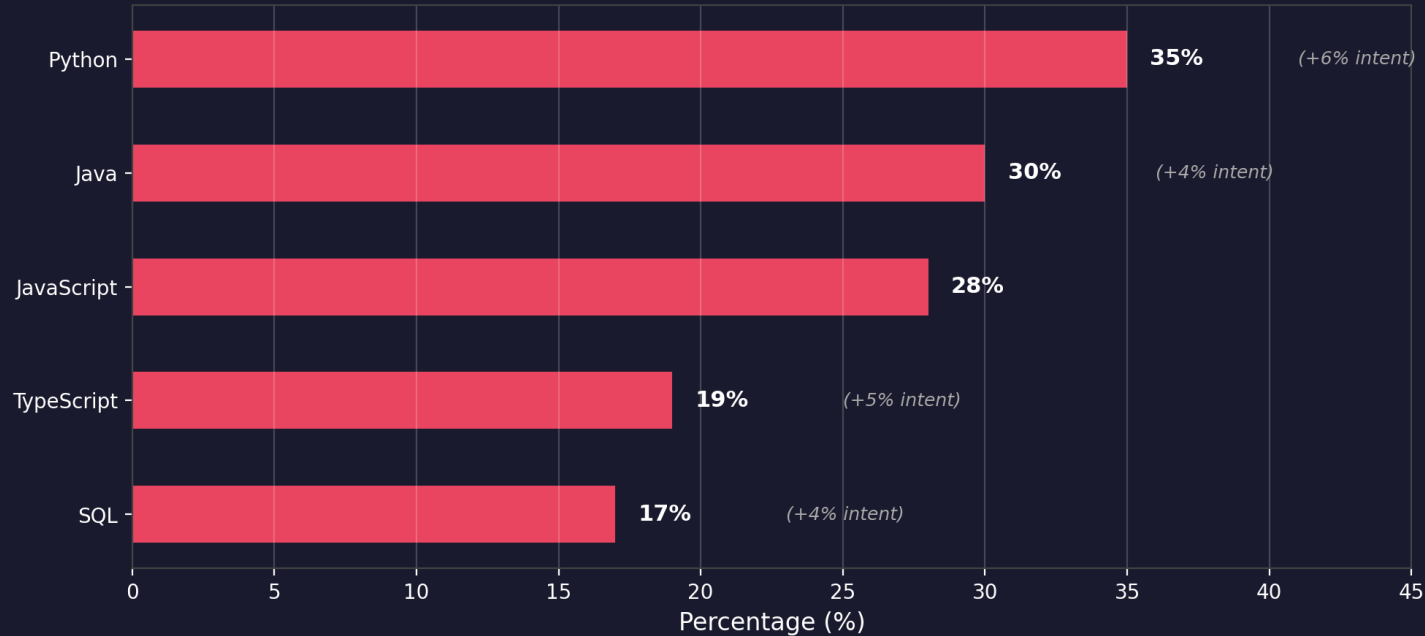
Programming Language Usage Trends (2017-2024)



Key Insight: JavaScript peaked at 70% (2020) and declined to 61%; Python rose from 32% to 57%; TypeScript tripled from 12% to 37%.

Python Is Now the #1 Primary Language at 35%

Primary Language Breakdown

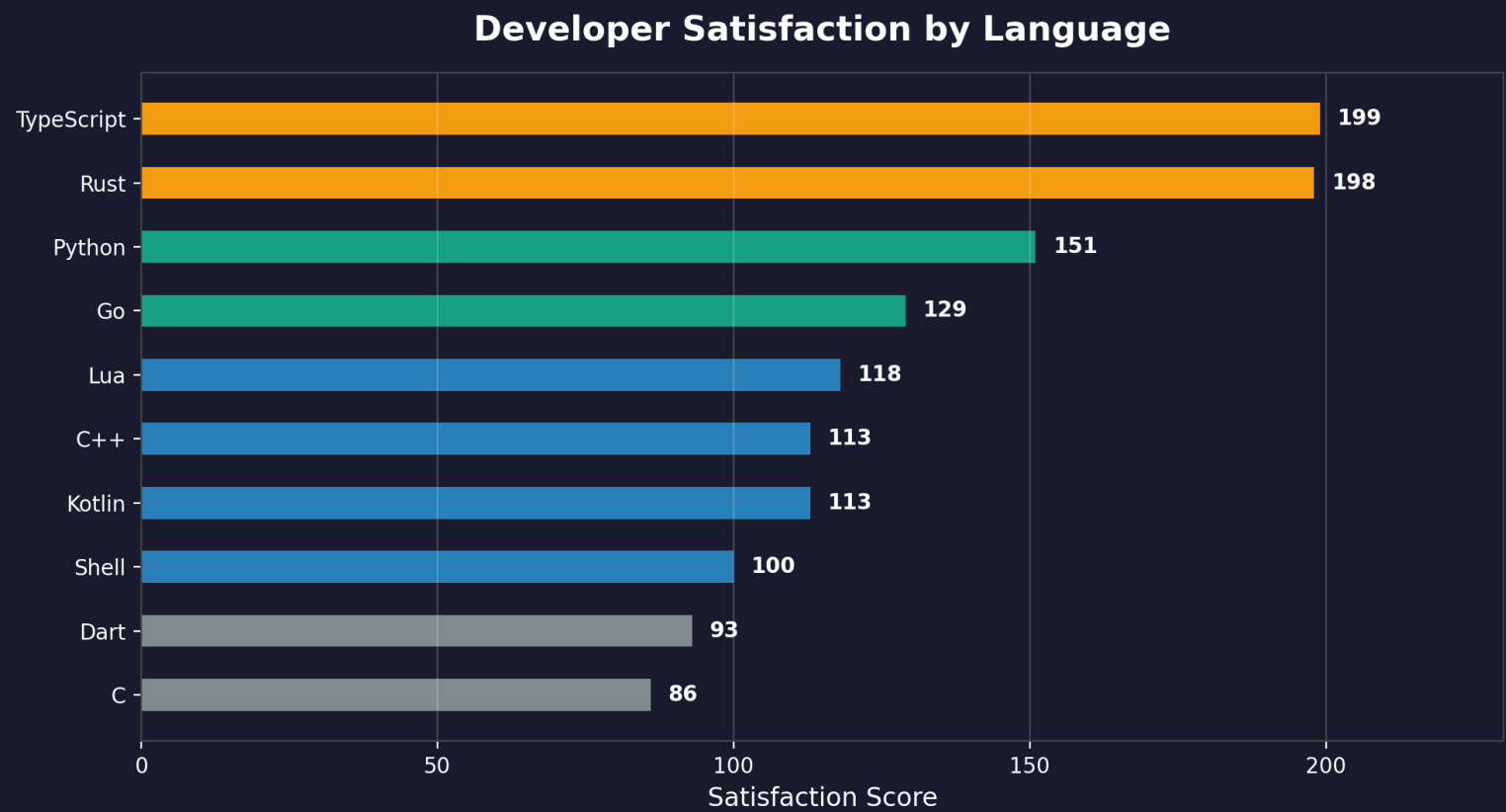


Ecosystem Lock-In Effect

- Python: 35% primary, only 6% adoption intent
- Java: 30% primary, only 4% adoption intent
- JavaScript: 28% primary
- TypeScript: 19% primary, 5% intent
- SQL: 17% primary, 4% intent

Adoption intent remains below 6% across all languages — developers stick with what they know.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



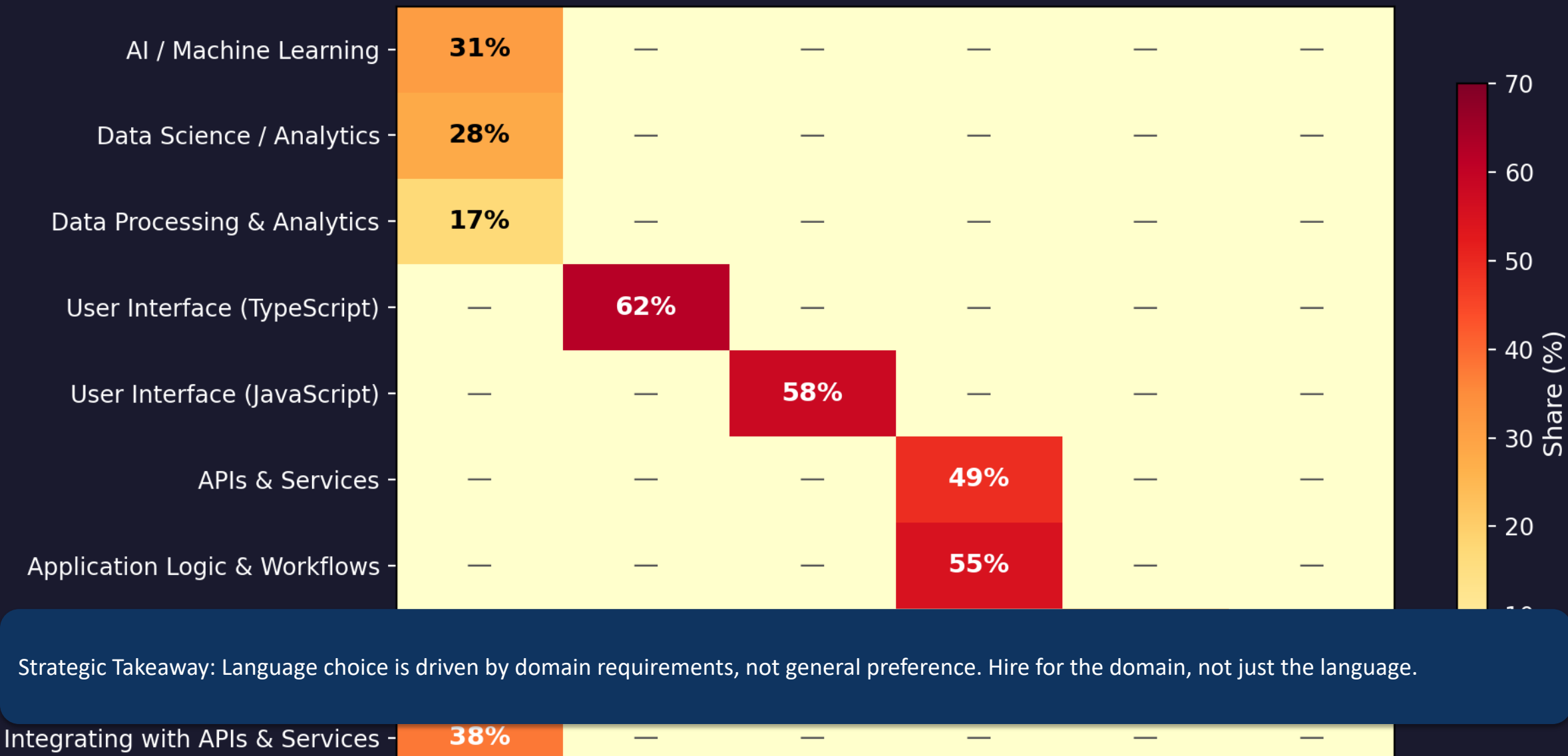
What Drives Satisfaction?

- Type safety and modern tooling dominate the top spots
- TypeScript (199) and Rust (198) are nearly tied
- Python (151) scores well due to ecosystem breadth
- Go (129) benefits from simplicity and concurrency

Languages with strong type systems and developer experience tools correlate with higher retention and team satisfaction.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language Dominance by Application Domain



What This Means for Your Technology Strategy in 2025

1 Invest in Python for AI/ML Teams

Python is the undisputed leader in AI/ML (31%), data science (28%), and data processing. It is now the #1 primary language at 35%.

2 TypeScript Is the Future of Frontend & Full-Stack

TypeScript tripled to 37% usage, leads UI development (62%), and tops satisfaction scores (199). The default choice for modern web.

3 Rust and Go Are Growth Plays for Infrastructure

Go stabilized at ~18% after rapid growth; Rust climbs to 11%. Both prioritize performance, concurrency, and compiler safety — ideal for cloud-native and systems work.

4 PHP and Ruby Hiring Pools Are Shrinking

PHP dropped from 30% to 17%; Ruby fell from 10% to 4%. Legacy maintenance costs will rise as talent supply contracts.

5 Factor Developer Satisfaction Into Retention

TypeScript and Rust's high satisfaction correlates with type safety and modern tooling. Investing in DX reduces churn and attracts top talent.